

Your grade: **100%**

Your latest: **100%** • Your highest: **100%** • To pass you need at least 50%. We keep your highest score.

Next item →

1.

What is the primary function of a convolutional layer in a CNN?

1 / 1 point
- ☒

To detect features such as edges and textures in an image

☐

To reduce the spatial dimensions of the input image

☐

To perform classification tasks based on extracted features

☐

To normalize the input data to a standard scale
- ☒

Correct

Correct! Convolutional layers are designed to detect different features in an image, such as edges, textures, and patterns.

2.

Which of the following best describes the role of zero-padding in a convolutional neural network?

1 / 1 point

☐

Zero-padding increases the depth of the input image.

☒

Zero-padding adds extra pixels around the border of an image to control the spatial dimensions of the output.

☐

Zero-padding normalizes the input image to have zero mean and unit variance.

☐

Zero-padding smoothenes the input image before processing.

☒

Correct

Correct! Zero-padding is used to add extra pixels around the border of an image, which helps control the spatial dimensions of the output feature map.

3.

Which of the following statements about padding in convolutional neural networks are true?

1 / 1 point

☒

Padding can help preserve the spatial dimensions of the input.

☒

Correct

Correct! Padding is often used to maintain the spatial dimensions of the input image after the convolution operation.

☐

Padding adds additional parameters to the model.

☒

Padding can be used to control the output size of the convolutional layer.

☒

Correct

Correct! Padding is used to influence the spatial dimensions of the output feature map.

☐

Padding reduces the computational cost of the convolutional layer.

4.

What is the primary purpose of using a convolutional neural network (CNN) in a multiclass image classifier?

1 / 1 point

☐

To handle missing data in the dataset

☐

To optimize the learning rate during training

☐

To reduce the dimensionality of the data

☒

To extract spatial features from images

☒

Correct

Correct! CNNs are specifically designed to capture spatial hierarchies in images, making them suitable for tasks like image classification.

5.

What is the role of the activation function 'ReLU' in a CNN layer?

1 / 1 point

☐

Normalizing the input features

☐

Reducing the number of parameters in the model

☒

Introducing non-linearity into the network

☐

Preventing overfitting during training

☒

Correct

Correct! ReLU helps in introducing non-linearity, allowing the network to learn complex patterns.